

BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE422 : Artificial Intelligence
Assignment 03

1. Consider the following probabilities for the events A, B, and C to solve the problems below:

$$P(A, B, C) = 0.2$$

$$P(A, \neg B, C) = 0.15$$

$$P(A, \neg B, \neg C) = 0.05$$

$$P(\neg A, B, C) = 0.05$$

$$P(\neg A, B, \neg C) = 0.05$$

$$P(\neg A, \neg B, C) = 0.35$$

$$P(A) = 0.5$$

$$P(\neg C) = 0.25$$

- a. Create a joint probability distribution table for the events.
- b. Calculate $P(\neg B)$.
- c. Calculate $P(A | \neg B, C)$ and $P(\neg A | B, \neg C)$.
- d. Determine whether A, B, and C are independent or not.

2. The following table presents data, used to predict course difficulty based on prior experience, learning goals and available time.

Prior Experience	Learning Goal	Availability	Course Difficulty
None	Explore	> 10 hours	Beginner
Extensive	Skill Improvement	5–10 hours	Advanced
None	Professional	< 5 hours	Beginner
Extensive	Professional	> 10 hours	Advanced
Some	Professional	5–10 hours	Intermediate
None	Skill Improvement	< 5 hours	Beginner
Some	Explore	> 10 hours	Intermediate
Extensive	Professional	< 5 hours	Intermediate
Some	Skill Improvement	> 10 hours	Intermediate
None	Explore	5–10 hours	Beginner
None	Explore	> 10 hours	Intermediate
Extensive	Professional	5-10 hours	Intermediate
Some	Professional	> 10 hours	Advanced
Some	Skill Improvement	5–10 hours	Beginner
Extensive	Professional	< 5 hours	Advanced

- a. Find the probability that a randomly selected student is taking an Advanced course.
- b. Find the joint probability that a student has no prior experience and is taking a beginner-level course.
- c. Find the probability that the course is Beginner given that a student has no prior experience.
- d. Using Naive Bayes, suggest the course difficulty for a student with some prior experience, a goal of professional certification, and more than 10 hours of available time.
- e. Explain why the denominator can be omitted when comparing probabilistic results and making decisions using Bayes' Theorem.
- f. Explain the zero-frequency problem in Naive Bayes classification with a suitable example.